

Short communication

Microstructure of bottlebrush poly(n-alkyl methacrylate)s beyond side chain packing

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ABSTRACT

Based on results of small and wide angle X-ray scattering measurements, we report a new microstructural feature in bottlebrush poly (n-alkyl methacrylate)s larger than side chain packing. Temperature dependent X-ray experiments confirmed that such large scale aggregation structure is originated from the crystallization of the side chains. Two-dimensional X-ray scattering patterns revealed that the structure is due to the formation of disordered regions locate on average two layers of well crystallized side chains apart. Crystallization of the side chains rejected backbone entanglements to the surrounding area forming regions of high entanglement density so that restricting the crystallization of side chains. As such a density contrast is created between the crystallized regions and the disordered ones which results in strong small angle X-ray scattering signals.

1. Introduction

Polymers with regular side chains attached to backbone are traditionally termed as comb-like polymers. However, recent molecular dynamics simulations by Liang et al. revealed that the crowding parameter defined by the ratio of length of side chain and spacing on backbone has obvious influence on architecture of macromolecules [1]. It is thus possible to specify bottlebrushes from common definition of comb-like polymers if the crowding parameter $\phi \geq 1$. Therefore, bottlebrushes are a type of particular graft-copolymers with relatively long side chains and high graft density. Their distinction from normal graft-copolymers gives rise to special aggregation structure and physical properties [2–4]. On account of the special aggregation structure, bottlebrushes with alkyl side chains are widely used as phase change materials, nanotemplate, photonic crystal, biological medicine and so on [5–10]. In general, bottlebrushes contain two categories: flexible backbone and rigid backbone bottlebrushes. The bottlebrushes with rigid backbone are widely investigated, because the graft degree and the length of alkyl side chains can adjust the melting temperature and solubility of polymers effectively [11–16].

Flexible backbone bottlebrushes have been investigated too since the

successful preparation of n-alkyl acrylates polymers [17]. The microstructure of Poly α -olefins (P α Os) were well investigated by Lopez-Barron et al. [18,19]. They reported that the whole bottlebrush syndiotactic P α Os and isotactic poly(1-octadecene) (POD) molecules pack into a hexagonal lattice, which in turn are bundled into fibrils that are spaced by a characteristic distance d_{fb} . Meanwhile, they compared the melt nonlinear viscoelastic response between POD and polypropylene and supposed that the strong strain hardening in POD is caused by the strain-induced enhancement in inter-chain friction drag. The friction is attributed to the partial overlapping and packing direction of side chains perpendicular to the flow direction [20]. Up to now, there are only a few investigations about the aggregation structure of flexible backbone bottlebrushes in solid state [20–22].

Poly (n-alkyl methacrylate)s (PMA(n)s) is a kind of typical flexible backbone bottlebrushes which has been mostly studied [11,23–32]. Long alkyl side chains tend to aggregate into small alkyl nano-domains, this aggregation phenomenon takes place not only for crystalline state alkyl side chains but also for alkyl side chains in amorphous state which can be observed by small angle X-ray scattering (SAXS) [33]. When the side chain length is longer than 12 carbons, PMA(n)s exhibit side chain crystallization behavior, and they form a layered structure. In the

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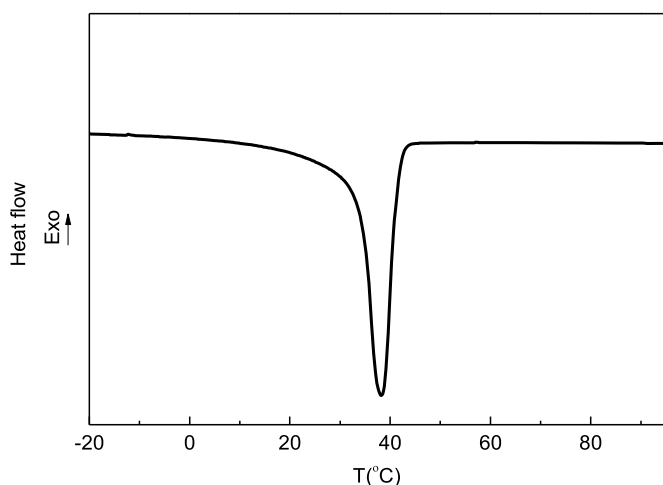


Fig. 1. DSC heating thermogram of PMA(18).

layered structure, backbone and crystallized side chains arrange alternately [25]. This result is demonstrated by wide angle X-ray diffraction (WAXD) and nuclear magnetic resonance (NMR) results [24,34]. Furthermore, when the side chains are in crystalline state, they are packed parallel to each other and perpendicular to the backbone, according to polarized infrared spectroscopy (IR) results [23]. It was concluded that the spacing of the layered structure in PMA(n)s is about few nanometers depending on the length of the side chains as can be justified by series of equal distanced scattering peaks in the large q side of SAXS and small q side of WAXD [23–26]. The behavior is considered to be much like the crystallization of paraffins having very regular lamellar crystals stacked on top of each other. However, as one end of the paraffin molecules is fixed by the backbone in PMA(n)s and the entanglements effect of the overall polymers, one would expect structural defects after side chain crystallization in such systems. It is thus the purpose of this work to investigate the structural characteristics of the PMA(n)s using X-ray scattering technique. As it turns out that we found an aggregation structure of PMA(n)s with a scale much larger than the paraffin packing. The results are interpreted by a model consisting dispersed amorphous regions apart from each other over two crystalline layers on average within a continuous crystalline matrix.

2. Experimental section

2.1. Materials

Stearyl methacrylate was purchased from Ailan Chemical Engineering Technology Co.. Hexadecyl methacrylate and docosyl methacrylate were supplied by Shanghai Dibai Chemicals Technology Co.. All monomers were directly employed for polymerization without further purification. Nano- CaCO_3 was provided by Ruicheng Huana

Nanomaterials Technology Co. and used as received. Chemical grade Benzoyl peroxide was offered by Sinopharm Chemical Reagents Co. and purified by recrystallization using trichloromethane as solvent and methanol as precipitant at room temperature.

2.2. Synthesis of poly (*n*-alkyl methacrylate)s (PMA(n)s)

A series of PMA(n)s with different side chain length were prepared. PMA(n)s were synthesized via conventional suspension polymerization. The brief preparation procedure was described as follows: Take PMA(18), the resultant polymer with a side chain length of 18 carbons, as an example. 1.0 g nano- CaCO_3 was evenly dispersed in 150 mL deionized water. Subsequently, 25 g stearyl methacrylate was added into the above dispersion under mechanical stirring at 50 °C for a while. Then 0.050 g initiator BPO was added into the reaction mixture with temperature increasing to 95 °C under inert condition (N_2 flow). After polymerization, particles of PMA(18) was filtrated and washed with dilute hydrochloric acid to remove the residual nano- CaCO_3 . Finally, PMA(18) was dried in an oven at 80 °C over 24 h. Typical size of the obtained PMA(18) particles ranges from 0.1–1 mm. The corresponding molecular weight of the obtained PMA(18) is 815.2 kg/mol with a polydispersity of 1.35 as determined by gel permeation chromatography. The materials were compression molded into sheet of 0.5 mm in thickness at 100 °C, then quenched in ice water. Oriented PMA(18) sample was obtained by stretching the compression molded sheet under 60 °C followed by quenching in ice water.

2.3. Characterizations

2.3.1. Differential scanning calorimetry (DSC)

A Mettler Toledo differential scanning calorimetry (mode DSC3/700/520) was used to study the thermal properties of PMA(n)s. The test condition of the DSC measurement was as follows: under N_2 flow (50 mL/min), 5–10 mg sample was encapsulated in an aluminum pan. The heating/cooling rate was 10 K min^{-1} and the temperature range was from –50 to 100 °C. The whole process was repeated twice. All experimental data were recorded from the second DSC thermograms.

2.3.2. WAXD

WAXD experiments were performed on a custom made system, which was equipped with a Cu $\text{K}\alpha$ X-ray source (GeniX3D Cu ULD, Xenocs, Sassenage, France, $\lambda = 0.154$ nm). A Pilatus 100 K detector (Dectris, Swiss) was used for collecting 2D WAXD diagrams. A heating stage was employed to ensure temperature control of the samples during measurements. The crystal melting process of PMA(18) was recorded via in-situ WAXD with a step-wise heating process from room temperature to 160 °C. Every WAXD diagram was acquired during a 5 min exposure time at a distance from sample to detector of 39 mm.

2.3.3. SAXS

SAXS measurements were performed to explore the aggregation state of PMA(n)s. The basic SAXS data of every sample was collected on a

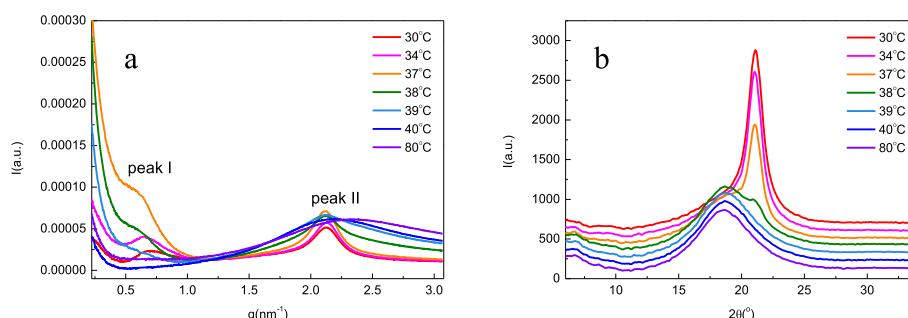


Fig. 2. Selected temperature dependent SAXS (a) and WAXD (b) profiles of PMA(18).

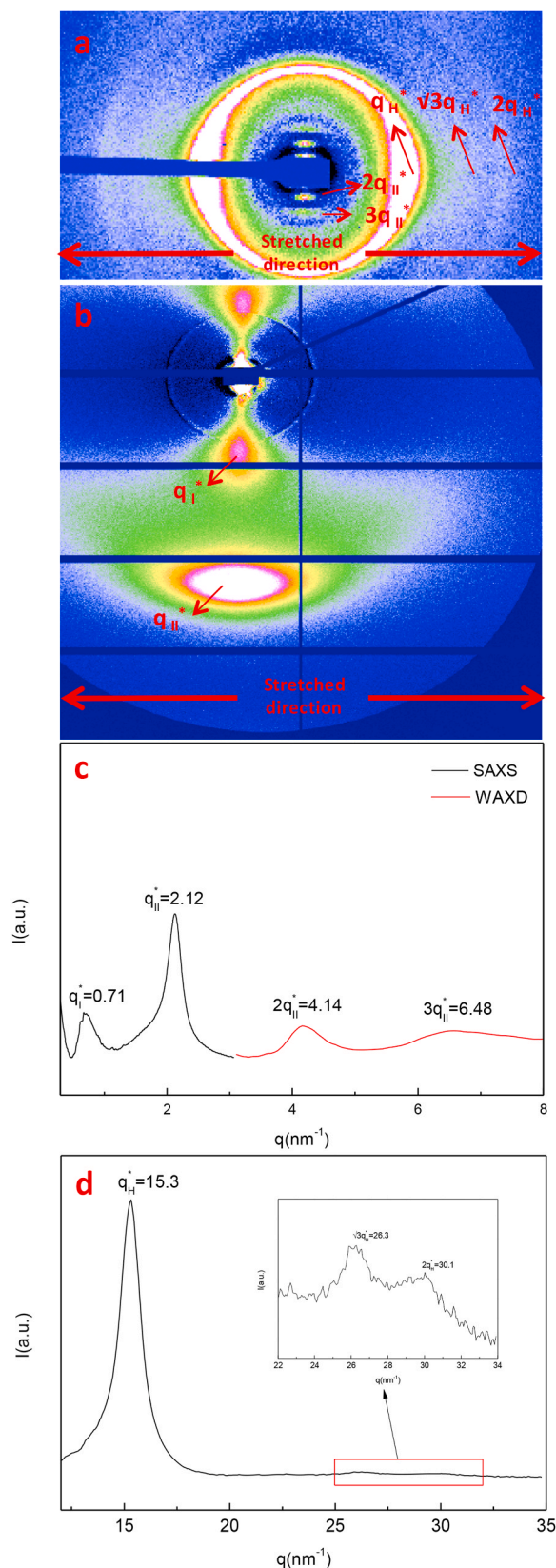


Fig. 3. 2D WAXD (a) and SAXS (b) patterns of oriented PMA(18) and the corresponding selected 1D X-ray scattering integral curves of the reflections being perpendicular (c) and parallel (d) to stretching direction; The circular intensity distribution close to Peak I is due to instrumental background.

modified Xeuss of Xenocs. Same detector and X-ray source as the WAXD set-up were used but a different focusing mirror producing near parallel beam and two sets of collimating scatterless slits were used in the SAXS set-up. The sample to detector distance was 1055 mm. Every sample was placed on thermal stage to ensure the testing temperature below the melting point of samples. At the same time, in order to obtain a more complete 2D diagram, the corresponding fraction of an in-situ SAXS experiment of PMA(18) and the acquisition of 2D SAXS pattern about oriented PMA(18) have been obtained at beamline 1W2A BSRF, Beijing, China ($\lambda = 0.154$ nm). The collection for every SAXS diagram was 5 min with the distance from sample to detector of about 1590 mm. The samples of PMA(n)s were mounted on a heating stage after being wrapped tightly in Kapton film. During the in-situ SAXS experiment, the temperature profile was the same as used in in-situ WAXD experiments.

3. Result and discussion

Fig. 1 depicts the data of DSC heating scan of PMA(18). The curve exhibits an obvious endothermic peak during the heating process illustrating that the side chains of PMA(18) could crystallize at suitable temperature. DSC heating scans of PMA(16) and PMA(22) have been shown in the Supporting Information documents as **Fig. S1**. It showed that with the increase of side chain length, the melting temperature increased from 24.7 °C to 55.6 °C, and the corresponding enthalpy also increased. The detailed thermal information was listed in **Table S1** in the Supporting Information document.

To investigate microstructures of the PMA(n)s, we performed in-situ temperature dependent SAXS and WAXD measurements for PMA(18) and presented some selected data in **Fig. 2**. Two scattering peaks were observed at different q positions in the SAXS curves taken at temperatures below the melting point of PMA(18) manifesting the existence of two kind of ordered/correlated structures with different length scales. In **Fig. S2** of the Supporting Information document, similar features have been also observed showing that such structural features are universal for PMA(n)s. The spacing of the two ordered structures (d_I and d_{II} in the order of decreasing) can be calculated according to the Bragg equation of $d = 2\pi/q_{\text{peak}}$, where q_{peak} is the intensity peak position on the scattering curve. d_I and d_{II} values for different PMA(n)s were summarized in **Table S1**. According to the previous literatures [28], peak II referred to the layered structure formed by the alternated side chains and backbones, and the corresponding d_{II} represented the distance between the neighboring backbones. d_{II} increased with the increase of side chain length as shown in **Table S1**. The side chain length of PMA(18) was 2.3 nm when extended. Therefore, the observed d_{II} of 3.0 nm in PMA(18) indicated that the side chains must be packed in a tilted or interdigitated way. At this stage, it was not possible to elucidate the exact side chain packing structure based on purely the isotropic scattering data. The assignment of the structural basis of d_I was also not straightforward based on the current isotropic scattering data. The WAXD profiles shown in **Fig. 2b** indicated that the side chains of PMA(18) may be packed in a hexagonal structure with one sharp diffraction peak at low temperature. With the increase of temperature, one observed a rather simultaneous change in the intensity distribution profiles in both SAXS and WAXD curves. All peaks (d_I and d_{II} in SAXS and the diffraction peak in WAXD) disappeared at high temperatures when the sample was molten leaving only two broad amorphous halos corresponding to the average backbone-backbone and side chain-side chain distances. Notably, the backbone-backbone distance became smaller after melting of the crystals as evidenced by the shifting of the corresponding peak position to large q values. This result suggests that upon crystallization of the side chains the backbones of adjacent bottlebrushes must be pushed away from each other. This result strongly suggests that all observed structural features are related to the side chain crystallization.

To better understand the actual micro-structure of the samples, an uniaxially oriented PMA(18) sample have been measured using 2D WAXD and SAXS techniques. The corresponding patterns and selected

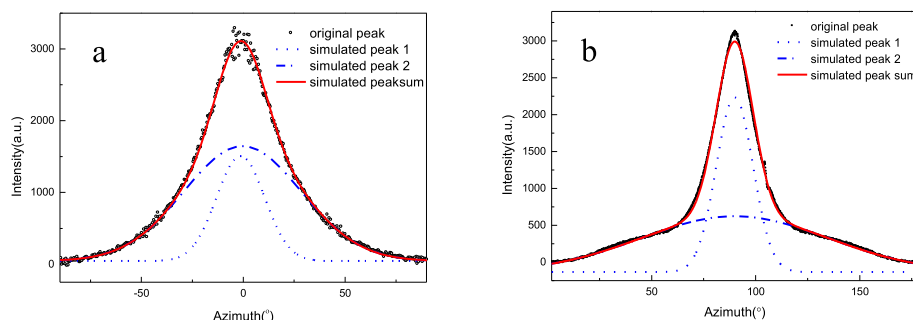


Fig. 4. Azimuthal integrated intensity curve of WAXD at 21.6° (a) and SAXS for Peak II (b) of oriented PMA(18). Reference direction (0°) is along the stretching direction.

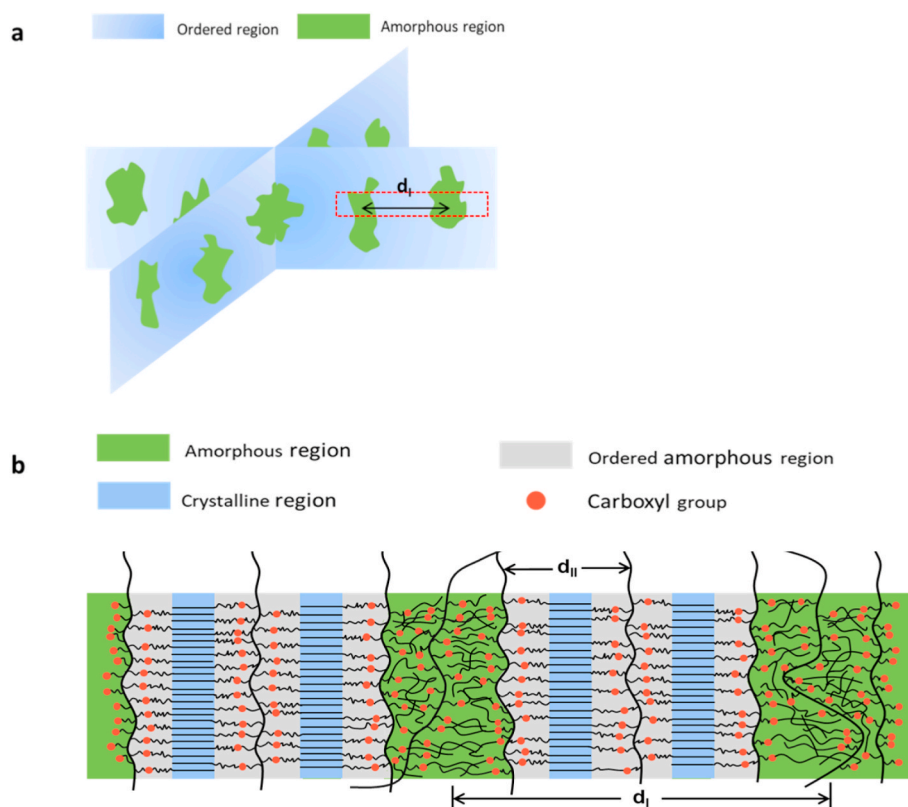


Fig. 5. Possible structure model of side chain crystallized PMA(n)s with section views (a) and enlarged local view (b).

1D integrated curves were shown in Fig. 3. Both WAXD and SAXS patterns show strong anisotropic scattering intensity distributions at q positions exactly same as in the isotropic sample indicating that there was only a reorientation of the same structure upon stretching. The X-ray patterns of the uniaxially oriented sample can help to reveal the actual chain packing modes of the PMA(n)s. Peak II in SAXS patterns located at the positions perpendicular to the stretching direction showing the preferential alignment of the backbones along the orientation direction. Moreover, as shown in Fig. 3c, selected 1D integrated curves of the scattering signals on both SAXS and WAXD patterns perpendicular to the stretching direction clearly give peaks at q^* , $2q^*$, $3q^*$. Therefore, the equidistant higher order scattering peaks of Peak II appeared in 2D WAXD pattern in the middle angle range manifesting the layer-like structure of backbone packing due to side chain crystallization. With the diffraction pattern of the oriented sample, it is also possible to confirm the hexagonal packing of the side chains upon crystallization. As was shown in Fig. 3a on the 2D pattern and Fig. 3d on the integrated 1D curve, additional diffraction peaks are clearly identified showing

exactly features of hexagonal packings.

It is interesting that the scattering intensity of Peak I also locates at the same direction indicating similar origin of this large scale structure. The diffraction intensities in WAXD pattern due to side chain packing are concentrated along the stretching direction. To illustrate the relative orientation of the backbones and side chains, azimuthal intensity distributions of the side chain diffraction in WAXD and the Peak II in SAXS are shown in Fig. 4. Both curves of the azimuthal intensity distributions in Fig. 4 can be decomposed into two components of different degree of orientation. This is because of the limited deformation ratio obtained during the stretching process. Clearly, the diffraction peak of the packed side chains and the oriented backbones both show single peak and locate at positions perpendicular to each other in the azimuthal intensity distribution curves. Such results against strongly possible tilted packing structure of side chains leaving the only possibility of interdigitated structure after side chain crystallization. Therefore, the side chains of PMA(18) were paralleled to each other and perpendicular to the backbones. This result was consistent with the previous work demonstrated

by Plate via polarized IR [23].

The appearance of Peak I in SAXS data suggests the existence of a structure much larger than the backbone-backbone packing distance. Similar SAXS peak observed in P α Os has been assigned to the interfibrillar distance where the fibrils are due to hexagonally packed bottlebrush chains [18,19]. However, since the PMA(n)s possess a very different layer-like packings compared to the P α Os, the observed Peak I in our SAXS curves must have its different origin. As was given in Table S1, the average period of the Peak I is nearly 3 times of Peak II. Given the fact that several higher order peaks of Peak II are clearly visible, the side chain crystallization must result in multiple layers of alternative backbone-backbone packings. Therefore, the observed Peak I should not be understood as a result of interrupting of the layer-like backbone packing every two layers on average. Rather, it could be originated from amorphous regions randomly but correlated embedded into the crystalline matrix. This result is somewhat surprising as it has been mostly considered that every side chain crystallizes in the PMA(n) bottlebrushes when the length of the side chain is larger than a critical value. Current findings of the super structure larger than the backbone packing distance however suggest a different situation. In order to provide sufficient electron density contrast to produce the Peak I in SAXS at all, the system must possess a fraction of side chains that are not crystallizable even at temperatures below melting point. This conclusion might be odd at the first sight as the side chains are so short which should be able to arrange themselves into certain packed structures. However, when the properties of backbones are considered, the observed behavior becomes reasonable. The bottlebrushes PMA(n)s used in this study possess high viscosity indicating a high molecular weight and highly entangled structure in the melt. Upon crystallization of the side chains, the entanglements of the backbones can only be shifted to the surrounding areas of the crystallites resulting in a concentration of the entanglements at certain locations within the already formed crystalline structure. These regions of high entanglement density are obviously unfavored for possible crystallization of the side chains because of the backbone conformation. Moreover, as indicated in Fig. 1a, when the side chains start to participate in crystallization, the distance between the backbones with crystallizable side chains become larger as discussed above. This together with enriched entanglements renders these regions free from side chain crystallization.

Based on the above analysis, we could propose a structural model as follows:

The global structure of the system was formed by the interdigitated side chain crystalline region including the connecting side chains between the backbones and crystallized side chains. The amorphous regions embed in the continuous crystalline matrix and the average distance of the amorphous regions is about two layers of backbone-backbone packings. A schematic illustration of the structural features of the PMA(n)s after crystallization is shown in Fig. 5. Obviously, this structure is very different from the ordinary semicrystalline lamellar structure where the backbones are oriented along the normal of the layer-like lamellae. The layered structure observed in the bottlebrushes is governed by the crystallization of side chains resulting in a perpendicular orientation of backbones with respect to the normal of the crystalline layers. The effect of the backbones on the microstructure is reflected by the existence of additional disordered layers due to enriched entanglements. Thus, the super-structure of a bottlebrushes is determined by synergistic effects of backbones and side chains together.

4. Conclusion

In conclusion, we have shown that a new microstructure larger than the side chain packing exists in a series of poly (n-alkyl methacrylate)s with n ranging from 16 to 22 where these n-alkyl side chains are crystallizable. Side chain crystallization in these bottlebrushes gives rise to a global layer-like structures with side chains being interdigitated packed hexagonally and perpendicular to the backbones. Due to the existence of

backbone entanglements and the requirement of expending of the backbone-backbone distance upon side chain crystallization, regions of amorphous nature located on average two crystalline layers apart from each other are embedded in the continuous crystalline matrix. The observed super structure beyond side chain packing in the bottlebrushes has its consequence in understanding their properties and further work in that direction is on the way.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.polymer.2020.123034>.

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